NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2.

SATURDAY 17TH JUNE, 2017 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 19th June, 2017**.

(d) **ALL** *participants who qualified and sat for Paper* 2 *are cordially invited to the certificate and/or prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*

SECTION A : 5 marks each

Qn. 1. Two positive numbers a and b satisfy $2+\log_2 a = 3+\log_3 b = \log_6(a+b)$. Compute the value of

$$\frac{1}{a}+\frac{1}{b}.$$

Qn. 2. Given the sequence $\{x_n\}$ defined by $x_{n+1} = \dfrac{1+x_n\sqrt{3}}{\sqrt{3}-x_n}$ with $x_1 = 1$, compute the value of $x_{2016} - x_{618}$.

Qn. 3. Let $f(x) = \dfrac{9^x}{9^x+3}$. Evaluate the sum $f\left(\dfrac{1}{1996}\right) + f\left(\dfrac{2}{1996}\right) + f\left(\dfrac{3}{1996}\right) + \cdots + f\left(\dfrac{1995}{1996}\right)$.

Qn. 4. Prove that $\dfrac{1}{1999} < \dfrac{1}{2}\cdot\dfrac{3}{4}\cdot\dfrac{5}{6}\cdots\dfrac{1997}{1998} < \dfrac{1}{44}$

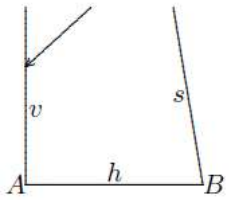
Qn. 5. You are trying to guess an integer between 1 and 1000, inclusive. Each time you make a guess, which must be an integer, you are told whether your number is too high, too low, or correct. What is the smallest number of guesses required to guarantee that you will have guessed the number?

Qn. 6. In triangle ABC, $BC = \sqrt{2}, CA = 6$, and angle $ACB = 135$ degrees. If C' is the midpoint of AB, what is the length of CC'?

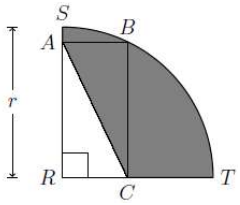
Qn. 7. Evaluate $\dfrac{1}{1\cdot 2\cdot 3} + \dfrac{1}{2\cdot 3\cdot 4} + \dfrac{1}{3\cdot 4\cdot 5} + \cdots + \dfrac{1}{98\cdot 99\cdot 100}$ as a simple fraction.

Qn. 8. Alan and Bill walk at 4 miles per hour, and Chad drives 40 miles per hour. They set out from the same point in the same direction with Alan walking, and Bill riding with Chad. After an hour, Bill gets out of the car and starts walking in the same direction they were going, while Chad turns the car around and drives back to pick up Alan. When the car gets back to Alan, Alan gets into the car, the car turns around and drives until it meets Bill. How many miles has Bill traveled when the car meets him? You may assume that no time is required to turn around or to change passengers.

Qn. 9. In the diagram below, line v is perpendicular to line h, and the angle at B is 75 degrees. A light ray has angle of reflection equal to angle of incidence. Our convention is to consider the angle which is ≤ 90 degrees. A ray has initial angle of incidence with v of 50 degrees. It follows a path in which it next hits wall h, then s, and then v again. What will be the number of degrees of its angle of incidence with wall v this (second) time?



Qn. 10. In the figure, arc SBT is one quarter of a circle with center R and radius r. The length plus the width of rectangle $ABCR$ is 8, and the perimeter of the shaded region is $10 + 3\pi$. Find the value of r.



SECTION B : 10 marks each

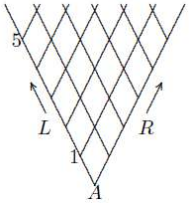
Qn. 11. Find all real numbers x such that $x = \left(x - \frac{1}{x}\right)^{1/2} + \left(1 - \frac{1}{x}\right)^{1/2}$.

Qn. 12. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(0) = 1$ and for any $x, y \in \mathbb{R}$ such that

$$f(xy + 1) = f(x)f(y) - f(y) - x + 2.$$

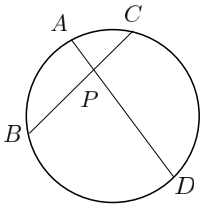
Determine $f(x)$.

Qn. 13. The bottom part of a network of roads is shown to the right. It extends up to level 99 in a similar fashion. 2^{100} people leave point A (level 0). At every intersection, half go in direction L and half in direction R. How many people will end up at position 1 at level 99?



Qn. 14. Let P be any point in the interior of a circle. Let two lines through P intersect the circle, one line, in points A and D, and another line, in points B and C as shown in the diagram. Prove that

$$AP \cdot PD = BP \cdot PC$$



Qn. 15. Without using a calculator:

(a) What integer equals $\sqrt{2000 \cdot 2008 \cdot 2009 \cdot 2017 + 1296}$?

(b) If

$$1.0000042376^2 = 1.00000xyz521795725376,$$

what is the value of $x + y + z$?

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SECTION A : 5 marks each

Qn. 1. Two positive numbers a and b satisfy $2+\log_2 a = 3+\log_3 b = \log_6(a+b)$. Compute the value of $\dfrac{1}{a}+\dfrac{1}{b}$.

Suppose

$$k = 2+\log_2 a = 3+\log_3 b = \log_6(a+b)$$

then we have $a = 2^{k-2}, b = 3^{k-3}, a+b = 6^k$, and therefore,

$$\frac{1}{a}+\frac{1}{b} = \frac{a+b}{ab} = \frac{6^k}{2^{k-2}3^{k-3}} = 2^2\times 3^3 = 108$$

Qn. 2. Given the sequence $\{x_n\}$ defined by $x_{n+1} = \dfrac{1+x_n\sqrt{3}}{\sqrt{3}-x_n}$ with $x_1 = 1$, compute the value of $x_{2016}-x_{618}$.

Note that $x_2 = 2+\sqrt{3}, x_3 = -2-\sqrt{3}, x_4 = -1, x_5 = -2+\sqrt{3}, x_6 = 2-\sqrt{3}, x_7 = 1 = x_1$, so that we have $x_{n+6} = x_n$ for all positive integers n; i.e., the sequence is periodic. Since

$$2016-618 = 1398 = 233\times 6,$$

we have $x_{2016}-x_{618} = 0$.

Qn. 3. Let $f(x) = \dfrac{9^x}{9^x + 3}$. Evaluate the sum

$$f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right) + f\left(\frac{3}{1996}\right) + \cdots + f\left(\frac{1995}{1996}\right)$$

Note that

$$f(1-x) = \frac{9^{1-x}}{9^{1-x}+3} = \frac{9}{9 + 3 \times 9^x} = \frac{3}{9^x + 3}$$

From which we get

$$f(x) + f(1-x) = \frac{9^x}{9^x+3} + \frac{3}{9^x+3} = 1$$

Therefore

$$\begin{aligned}
\sum_{k=1}^{1995} f\left(\frac{k}{1996}\right) &= \sum_{k=1}^{997}\left[f\left(\frac{k}{1996}\right) + f\left(\frac{1996-k}{1996}\right)\right] + f\left(\frac{998}{1996}\right) \\
&= \sum_{k=1}^{997}\left[f\left(\frac{k}{1996}\right) + f\left(1 - \frac{k}{1996}\right)\right] + f\left(\frac{1}{2}\right) \\
&= 997 + \frac{3}{3+3} \\
&= 997\frac{1}{2}
\end{aligned}$$

Qn. 4. Prove that

$$\frac{1}{1999} < \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{1997}{1998} < \frac{1}{44}$$

Let $P = \dfrac{1}{2} \cdot \dfrac{3}{4} \cdot \dfrac{5}{6} \cdots \dfrac{1997}{1998}$. But $\dfrac{1}{2} > \dfrac{1}{3}$ since $2 < 3$, $\dfrac{3}{4} > \dfrac{3}{5}$ since $4 < 5, \ldots, \ldots, \dfrac{1997}{1998} > \dfrac{1997}{1999}$ because $1998 < 1999$.

So

$$P > \frac{1}{3} \cdot \frac{3}{5} \cdots \frac{1997}{1999} = \frac{1}{1999} \tag{1.1}$$

Also $\dfrac{1}{2} < \dfrac{2}{3}$ because $1 \cdot 3 < 2 \cdot 2$, and $\dfrac{3}{4} < \dfrac{4}{5}$ because $3 \cdot 5 < 4 \cdot 4, \ldots$ $\dfrac{1997}{1998} < \dfrac{1998}{1999}$ because $1997 \cdot 1999 = 1998^2 - 1 < 1998^2$.

So

$$\begin{aligned}
P < \frac{2}{3} \cdot 4 \cdot 5 \cdots \frac{1998}{1999} &= \left[\frac{2}{1} \cdot \frac{4}{3} \cdot \frac{6}{5} \cdots \frac{1998}{1997}\right] \frac{1}{1999} \\
&= \frac{1}{P}\frac{1}{1999}
\end{aligned}$$

Hence

$$P^2 < \frac{1}{1999} < \frac{1}{1936} = \frac{1}{44^2} \quad \text{and} \quad P < \frac{1}{44} \tag{1.2}$$

Then (1.1) and (1.2) give

$$\frac{1}{1999} < P < \frac{1}{44}$$

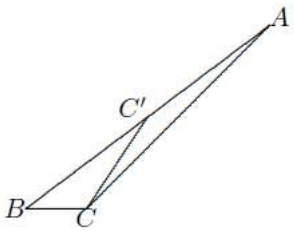
Qn. 5. You are trying to guess an integer between 1 and 1000, inclusive. Each time you make a guess, which must be an integer, you are told whether your number is too high, too low, or correct. What is the smallest number of guesses required to guarantee that you will have guessed the number?

They must be 10 guesses. This is because $2^{10} = 1024 > 1000$: The best that you can guarantee after 1 guess is to limit it to 500 numbers, then $250, 125, 62, 31, 15, 7, 3$, and 1. After 9 guesses, you know what the answer is, but you still must make the tenth guess.

Qn. 6. In triangle ABC, $BC = \sqrt{2}, CA = 6$, and angle $ACB = 135$ degrees. If C' is the midpoint of AB, what is the length of CC'?

Place B at $(0, 0)$, C at $\left(\sqrt{2}, 0\right)$ and A at $\left(4\sqrt{2}, 3\sqrt{2}\right)$. Then C' is at $\left(2\sqrt{2}, \frac{3}{2}\sqrt{2}\right)$ and so the desired length is

$$\sqrt{\left(\sqrt{2}\right)^2 + \left(\frac{3}{2}\sqrt{2}\right)^2} = \sqrt{2}\sqrt{1 + \frac{9}{4}} = \frac{\sqrt{26}}{2}$$



Qn. 7. Evaluate $\frac{1}{1\cdot 2\cdot 3}+\frac{1}{2\cdot 3\cdot 4}+\frac{1}{3\cdot 4\cdot 5}+\cdots+\frac{1}{98\cdot 99\cdot 100}$ as a simple fraction.

Note that

$$\frac{1}{(n+1)n(n-1)}=\frac{1}{2n(n-1)}-\frac{1}{2(n+1)n}$$

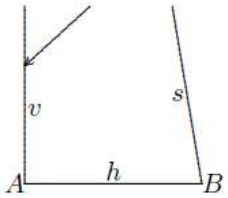
Thus the desired sum is $\frac{1}{4}-\frac{1}{2(100)(99)}$, as all other intermediate terms cancel out. This equals

$$\frac{50\cdot 99-1}{200\cdot 99}.$$

Qn. 8. Alan and Bill walk at 4 miles per hour, and Chad drives 40 miles per hour. They set out from the same point in the same direction with Alan walking, and Bill riding with Chad. After an hour, Bill gets out of the car and starts walking in the same direction they were going, while Chad turns the car around and drives back to pick up Alan. When the car gets back to Alan, Alan gets into the car, the car turns around and drives until it meets Bill. How many miles has Bill traveled when the car meets him? You may assume that no time is required to turn around or to change passengers.

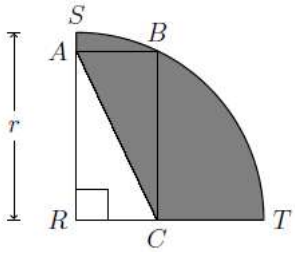
Let t denote the number of hours during which Chad was driving back to pick up Alan. Then $4(1+t)=40-40t$, hence $t=\frac{9}{11}$. Although you can compute it by solving equations, you can also reason it out that Bill will have to drive forward the second time for exactly one hour in order to catch up with Bill. The reason for this is that if we forget about the time period when they were both walking, we see that when they eventually meet, they both will have been in the car the same amount of time. So, during the two hours when one was walking and one riding each will have traveled 44 miles, and during the other 9/11 hour, each will have traveled 36/11 miles. Thus $47\frac{3}{11}$ and $\frac{520}{11}$

Qn. 9. In the diagram below, line v is perpendicular to line h, and the angle at B is 75 degrees. A light ray has angle of reflection equal to angle of incidence. Our convention is to consider the angle which is ≤ 90 degrees. A ray has initial angle of incidence with v of 50 degrees. It follows a path in which it next hits wall h, then s, and then v again. What will be the number of degrees of its angle of incidence with wall v this (second) time?



If it hits h at C, the angle there is 40 degrees. Then it hits s at point D with angle $180 - 75 - 40 = 65$. If it next hits v at point E, then $ABDE$ is a quadrilateral with three of its angles $90, 75$, and $(180 - 65)$. Since a quadrilateral has 360 degrees, this makes the desired angle $360 - 280 = 80$.

Qn. 10. In the figure below, arc SBT is one quarter of a circle with center R and radius r. The length plus the width of rectangle $ABCR$ is 8, and the perimeter of the shaded region is $10 + 3\pi$. Find the value of r.



Set $x = RC$ and $y = AR$. It must then be $x + y = 8$ (by hypothesis) and $x^2 + y^2 = r^2$ (since the rectangle $ABCR$ has one vertex in the circle). The perimeter of the shaded area is then

$$AS + SBT + TC + AC = (r - y) + r\frac{\pi}{2} + (r - x) + \sqrt{x^2 + y^2} = 3r + r\frac{\pi}{2} - (x + y) = 3r + r\frac{\pi}{2} - 8.$$

All we need to do is solve for r in the equation

$$18 + 3\pi = \left(3 + \frac{\pi}{2}\right) r.$$

SECTION B : 10 **marks each**

Qn. 11. Find all real numbers x such that

$$x = \left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}}$$

Since $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} \geq 0$ and $\left(1 - \frac{1}{x}\right)^{\frac{1}{2}} \geq 0$, then $0 \leq \left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$. Note that $x \neq 0$. Else, $\frac{1}{x}$ would not be defined, so $x > 0$. Squaring both sides gives,

$$\begin{aligned} x^2 &= \left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} + 2\sqrt{\left(x - \frac{1}{x}\right) + \left(1 - \frac{1}{x}\right)} \\ x^2 &= x + 1 - \frac{2}{x} + 2\sqrt{x - 1 - \frac{1}{x} + \frac{1}{x^2}}. \end{aligned}$$

Multiplying both sides by x and rearranging, we get

$$\begin{aligned} x^3 - x^2 - x + 2 &= 2\sqrt{x^3 - x^2 - x + 1} \\ \left(x^3 - x^2 - x + 1\right) - 2\sqrt{x^3 - x^2 - x + 1} + 1 &= 0 \\ \left(\sqrt{x^3 - x^2 - x + 1} - 1\right)^2 &= 0 \\ \sqrt{x^3 - x^2 - x + 1} &= 1 \\ x^3 - x^2 - x + 1 &= 1 \\ x(x^2 - x - 1) &= 0 \\ x^2 - x - 1 &= 0 \quad \text{since } x \neq 0 \end{aligned}$$

Thus $x = \frac{1 \pm \sqrt{5}}{2}$. We must check to see whether theses are indeed solutions.

Let $\alpha = \frac{1 + \sqrt{5}}{2}$, $\beta = \frac{1 - \sqrt{5}}{2}$. Note that $\alpha + \beta = 1$, $\alpha\beta = -1$ and $\alpha > 0 > \beta$.

Since $\beta < 0, \beta$ is not a solution.

Now, if $x = \alpha$, then

$$\begin{aligned} \left(\alpha - \frac{1}{\alpha}\right)^{1/2} + \left(1 - \frac{1}{\alpha}\right)^{1/2} &= (\alpha + \beta)^{1/2} + (1 + \beta)^{1/2} \quad \text{since } \alpha\beta = -1 \\ &= 1^{1/2} + (\beta^2)^{1/2} \quad \text{since } \alpha + \beta = 1, \text{ and } \beta^2 = \beta + 1 \\ &= 1 - \beta \quad \text{since } \beta < 0 \\ &= \alpha \quad \text{since } \alpha + \beta = 1 \end{aligned}$$

So $x = \alpha = \frac{1 + \sqrt{5}}{2}$ is a unique solution to the equation.

Qn. 12. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(0) = 1$ and for any $x, y \in \mathbb{R}$ such that

$$f(xy + 1) = f(x)f(y) - f(y) - x + 2.$$

Determine $f(x)$.

Since for $x, y \in \mathbb{R}$,

$$f(xy + 1) = f(x)f(y) - f(y) - x + 2 \tag{1.3}$$

We have

$$f(xy + 1) = f(y)f(x) - f(x) - y + 2 \tag{1.4}$$

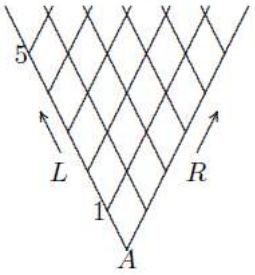
From (1.3) and (1.4)

$$\begin{aligned} f(x)f(y) - f(y) - x + 2 &= f(y)f(x) - f(x) - y + 2 \\ f(x) + y &= f(y) + x \end{aligned}$$

For $y = 0$, we obtain

$$f(x) = x + 1$$

Qn. 13. The bottom part of a network of roads is shown below. It extends up to level 99 in a similar fashion. 2^{100} people leave point A (level 0). At every intersection, half go in direction L and half in direction R. How many people will end up at position 1 at level 99?



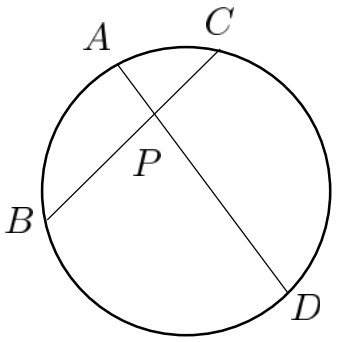
This is like Pascal's triangle. At level n, the fraction of the people in horizontal position i is $\frac{\binom{n}{i}}{2^i}$, and so the number of people in position 1 at level 99 is

$$\frac{\binom{99}{1}2^{100}}{2^{99}} = 198$$

Some students might use a more lengthy way of drawings and listings.

Qn. 14. Let P be any point in the interior of a circle. Let two lines through P intersect the circle, one line, in points A and D, and another line, in points B and C as shown in the diagram. Prove that

$$AP \cdot PD = BP \cdot PC$$



Follows from similarities of triangles ABP and CDP, a consequence of the equality of their angles:

- $\angle BAD = \angle BCD$ being angles subtended by same chord BD.
- $\angle ABC = \angle ADC$ being angles subtended by same chord AC.

$$\angle APB = \angle CPD \quad - \quad \text{clear}$$

So from similarity of triangles, ABP and CDP:

$$\frac{AP}{CP} = \frac{BP}{DP}\ , \quad \text{hence} \quad AP \cdot PD = BP \cdot PC$$

Qn. 15. Without using a calculator:

(a) What integer equals $\sqrt{2000 \cdot 2008 \cdot 2009 \cdot 2017 + 1296}$?

Let $A = 2000$ and $B = A^2 + 17A + 36$. The expression inside the square root is

$$A(A{+}17)(A{+}8)(A{+}9){+}1296 = (A^2{+}17A)(A^2{+}17A{+}72){+}1296 = (B{-}36)(B{+}36){+}1296 = B^2.$$

Thus the answer is $B = 20002 + 34000 + 36 = 4034036$.

(b) If

$$1.0000042376^2 = 1.00000xyz521795725376,$$

what is the value of $x + y + z$?

We know that $(1 + t)^2 = 1 + 2t + t^2$. Here $t = .0000042376$. The t^2 is much too small to affect the first three nonzero digits of $2t$. We obtain $(x, y, z) = (8, 4, 7)$. Therefore $x + y + z = 19$